Question 3 (12 Points)

In this question we consider the following point operation $M : [0, 1] \to [0, 1]$:

```python
def M(v, k1, k2):
    """Apply the point operation to a single gray level v."""
    if not 0 <= k1 < k2 <= 1:
        raise ValueError("Expected 0 <= k1 < k2 <= 1.")
    if v <= k1:
        return 0
    if v >= k2:
        return 1
    return (v - k1) / (k2 - k1)
```

Let f be a grayscale image with values in $[0, 1]$, and let $g(x) = M(f(x))$ be the image obtained by applying M pixelwise.

1. Invariance

Does there exist an image f with at least three distinct gray levels that is invariant under M (i.e., $M(f) = f$ pointwise)? If yes, give an example and prove it. If not, prove that no such image exists.

Answer

Yes. When $k_1 = 0$ and $k_2 = 1$, M is the identity operation:

- $v = 0$: $M(0) = 0$.
- $0 < v < 1$: $M(v) = \frac{v-0}{1-0} = v$.
- $v = 1$: $M(1) = 1$.

Thus, $M(f) = f$ for any image, including one with at least three distinct gray levels.

2. Mean intensity

Let $\mu(f)$ denote the mean gray level of f. How does $\mu(g)$ compare to $\mu(f)$? Is it always larger, always smaller, or dependent on f (and on k_1, k_2)? Explain.

Answer

The mean depends on f, k_1, and k_2. For example, with $k_1 = 0.4$ and $k_2 = 0.7$:

- Decrease: If $f \equiv 0.3$, then $\mu(f) = 0.3 > \mu(g) = 0$.
- Increase: If $f \equiv 0.75$, then $\mu(f) = 0.75 < \mu(g) = 1$.

3. Contrast & iteration

Assume for the sake of this section that f contains values only in $[k_1, k_2]$, such that $g \neq f$.

(a) Contrast

Compare $\mathrm{var}(g)$ and $\mathrm{var}(f)$ (contrast). Is $\mathrm{var}(g)$ always larger, always smaller, or dependent on f? Justify.

Answer

dependent on f, larger or equal.

Since all values of f lie in $[k_1, k_2]$,

$$g = \frac{f - k_1}{k_2 - k_1}, \qquad \mu(g) = \frac{\mu(f) - k_1}{k_2 - k_1}.$$

Subtracting the mean gives

$$g - \mu(g) = \frac{f - k_1}{k_2 - k_1} - \frac{\mu(f) - k_1}{k_2 - k_1} = \frac{f - \mu(f)}{k_2 - k_1}.$$

Therefore,

$$\begin{aligned} \mathrm{var}(g) &= \mathbb{E}\big[(g - \mu(g))^2\big] \\ &= \mathbb{E}\left[\left(\frac{f - \mu(f)}{k_2 - k_1}\right)^2\right] \\ &= \frac{1}{(k_2 - k_1)^2}\,\mathbb{E}\big[(f - \mu(f))^2\big] \\ &= \frac{\mathrm{var}(f)}{(k_2 - k_1)^2}. \end{aligned}$$

Since $0 < k_2 - k_1 \leq 1$,

$$\frac{1}{(k_2 - k_1)^2} \geq 1 \quad \Longrightarrow \quad \mathrm{var}(g) \geq \mathrm{var}(f).$$

For $k_1 = 0.2$ and $k_2 = 0.7$:

- **Constant image:** If $f \equiv 0.3$, then $g \equiv 0.2$. Both variances are zero.
- **Nonconstant image:** The variance strictly increases.

For equally frequent levels $f = \{0.3, 0.45, 0.6\}$, we get $g = \{0.2, 0.5, 0.8\}$, so $\mathrm{var}(g) = 4\,\mathrm{var}(f)$.

(b) Iteration

Consider repeated application $f^{(t+1)} = M(f^{(t)})$ with $f^{(0)} = f$. Describe the limiting image as $t \to \infty$.

Answer

The threshold is the fixed point satisfying

$$M(\theta) = \theta \quad \Longrightarrow \quad \frac{\theta - k_1}{k_2 - k_1} = \theta \quad \Longrightarrow \quad \theta = \frac{k_1}{1 - k_2 + k_1}.$$

Repeated application converges to:

$$M_\infty(v) = \begin{cases} 0, & v < \theta, \\ \theta, & v = \theta, \\ 1, & v > \theta. \end{cases}$$

Thus, pixels converge to 0 or 1, except those exactly at θ, which remain unchanged.

4. Negative after application

Does there exist an image h such that $M(h) = 1 - h$? If yes—construct and prove. If not—prove impossibility.

Answer

Yes. Choose a constant image $h \equiv v$, where

$$M(v) = 1 - v \quad \Longrightarrow \quad \frac{v - k_1}{k_2 - k_1} = 1 - v \quad \Longrightarrow \quad v = \frac{k_2}{1 + k_2 - k_1}.$$

This value lies in (k_1, k_2), so the middle branch applies and $M(h) = 1 - h$ pointwise.

Example: For $k_1 = 0.2$, $k_2 = 0.8$, and $h \equiv 0.5$,

$$M(h) = \frac{0.5 - 0.2}{0.8 - 0.2} = 0.5 = 1 - h.$$

```
k1 =0.1
k2 = 0.6
v= k2/(1+k2-k1)
print(f"v = {v:.3f}, k1 = {k1:.3f}, k2 = {k2:.3f}")
print(f"M(v, k1, k2) = {M(v, k1, k2):.3f}")
```

```
v = 0.400, k1 = 0.100, k2 = 0.600
M(v, k1, k2) = 0.600
```

Question 4 (8 Points)

Let I_1 be a grayscale image. First apply histogram equalization to obtain I_2. Then apply a point operation on I_2 to obtain I_3.

1. Overall point operation

Prove that the overall mapping from I_1 to I_3 is itself a point operation. That is, there exists a function M on gray levels such that $I_3(x) = M(I_1(x))$ for all pixels x.

Answer

For the given image I_1, histogram equalization defines a gray level mapping M_1, so

$$I_2(x) = M_1(I_1(x)).$$

Since the second operation is a point operation M_2,

$$I_3(x) = M_2(I_2(x)) = M_2(M_1(I_1(x))).$$

Define $M = M_2 \circ M_1$, meaning $M(v) = M_2(M_1(v))$. Then

$$\boxed{I_3(x) = M(I_1(x)) \quad \text{for every pixel } x.}$$

Thus, the output at each pixel depends only on its input gray level through the same function M, proving that the overall mapping is a point operation.

2. Uniform quantization and qualitative appearance

Answer

The mapping M uses the histogram equalization function computed from I_1, followed by the quantizer Q:

$$J_3(x) = Q(H_{I_1}(J_1(x)))\,.$$

The histogram of I_1 is concentrated at low gray levels, while J_1 contains mostly much brighter pixels. Consequently, the cumulative histogram of I_1 is already close to 1 at the gray levels appearing in J_1. Applying H_{I_1} therefore sends almost all pixels of J_1 to values near 255.

The quantizer has **bin width 64**, so it maps every intensity in $[192, 255]$ to 192. In particular,

$$Q(255) = \left\lfloor \frac{255}{64} \right\rfloor \cdot 64 = 192.$$

Thus, J_3 **will be essentially a uniform light gray image at intensity** 192, with its details disappearing.